

Late MoE is the real contribution.

Naive force is a straw improvement. Grade HIGH for Alt contact-rich syllabus; narrow on embodiment and open-world. Soft: multi-task USB 10%, unstable socket, no $\pi_{0.5}$.

Tab.3 0% → 80%

60.5 vs 37.3 / 40.2

Occlusion 90%

USB multi 10%

(b) **Visual Occlusion**, where parts of the plug and socket are hidden, introducing physical instability via clutter beneath the plug, testing the model's robustness and physical adaptability, with results shown in Table 3.

For generalization, particularly in scenarios requiring fine-tuned force control, ForceVLA achieved an 80.00% success rate, outperforming baseline VLMs. In the Height Gen. setting, ForceVLA demonstrated robustness by avoiding torque limit violations seen in baseline VLMs. Under visual degradation, ForceVLA maintained high success (90.00%) by leveraging multimodal feedback beyond visual cues. These results validate the ForceVLA architecture in intelligently integrating force feedback to modulate action in response to dynamic physical environments for robotic manipulation.

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Model	Success Rate
baseline [10]	45%
linear before VLM	55%
MoE before VLM	0
concat after VLM	60%
ForceVLA (Ours)	80%

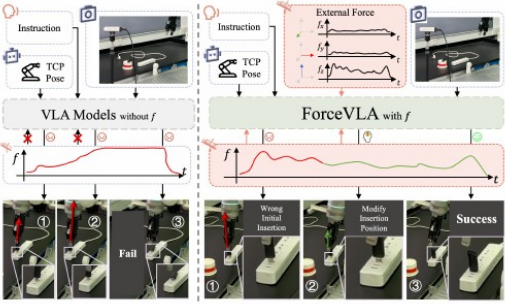


Figure 1: Comparison between ForceVLA and baselines without force input. Without force feedback, the policy fails to correct pose errors and completes insertion incorrectly. In contrast, ForceVLA leverages external force signals to adjust insertion strategies dynamically, leading to successful

Five facts that do not change with the seat

Late 6-axis F/T + FVLMoE

01

After VLM (PaliGemma/SigLIP): project wrench → force token → FVLMoE (E=4, top-1) → additive guide into π_0 flow-matching action head. Force stays late.

Contact-rich on Flexiv

02

Insert USB / plug, pump bottle, wipe board, peel cucumber. ForceVLA-Data: 244 traj / ~140k steps / ~50 demos per task. Quest3 teleop.

Must-cites

03

Avg 60.5 vs π_0 -base w/o F 37.3 (+23.2). Naive w/ F only 40.2 (+2.9). Early MoE-before-VLM 0% → late FVLMoE 80% (Tab.3). Occlusion 90%. FAST+naive force 31.0→14.2.

CLEAR contrasts

04

NOT MoPA-PD ancestor (MoPA = MP privilege→vision student; ForceVLA = deploy force into VLA). NOT JEPA/WM child. No $\pi_{0.5}$ bake-off — do not invent.

Number hygiene

05

Typo fix: 37.3 = w/o F (not w/ F). Preserve paper numbers. No invented BEHAVIOR SR. Pump caveat: naive w/F 83 > ForceVLA 67 — do not overclaim.

Five claims. Grade each, not the abstract alone.

C1**Force + FVLMoE $\gg$ π_0 baselines on contact-rich SR**

HOLDS — 60.5 vs 37.3/40.2; plug 80%

C2**Late fusion $\gg$ early into pretrained VLM**

HOLDS — Tab.3 MoE-before 0% vs ForceVLA 80%

C3**Naive concat insufficient vs MoE fusion**

HOLDS on avg (+2.9 vs +23.2); soft on Pump

C4**Generalizes under object/height/occlusion/unstable**

PARTIAL — occlusion 90%; unstable not best

C5**Router learns phase-aware specialization**

INTERPRETIVE — Fig.9; Expert-0 generalist is hypothesis

Early MoE destroys the VLM. Late FVLMoE wins.

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Table 3: Ablation Results	
Model	Success Rate
baseline10	45%
linear before VLM	55%
MoE before VLM	0
concat after VLM	60%
ForceVLA (Ours)	80%

Causal knife

MoE before VLM → 0%

ForceVLA (late) → 80%

Same late stage: concat 60 → FVLMoE 80 (+20 pp).
Early linear 55%; baseline 45%.

Softness

Task identity / trial N for Tab.3 underspecified vs main protocol.
Numeric match to Insert Plug is correspondence, not a stated
identity.

Authors' rule

Introduce force AFTER VLM. How you fuse beats presence.

Large gap. Force presence is not enough.

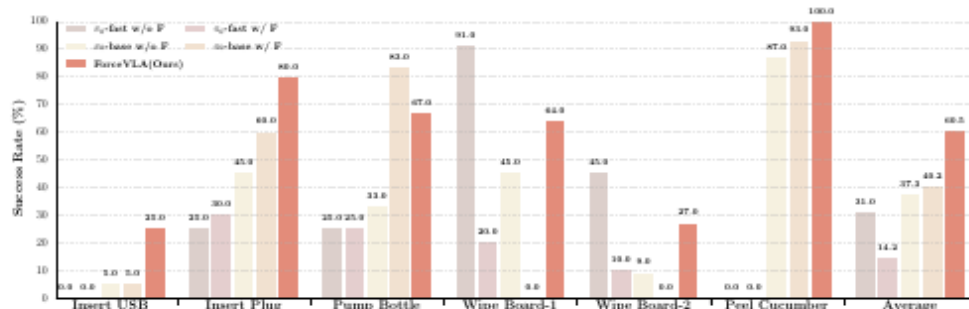


Figure 5: Main task success rates across different methods. ForceVLA significantly outperforms all baselines on five contact-rich tasks. Incorporating external force feedback improves performance for π_0 -base model, while our method achieves the highest average success rate, demonstrating robust performance under complex interaction dynamics. “Wipe Board-1” indicates the success rate of successfully performing the wiping motion, while “Wipe Board-2” refers to the success rate of completely erasing the markings.

ForceVLA shows an improvement of 23.2%. This highlights the substantial benefit of incorporating and effectively processing multimodal information, including force. Table 1 further highlights ForceVLA’s superior performance on the intricate *cucumber peeling* task. Our model excelled on both key metrics: it achieved the longest average peel length per stroke (14.12 cm $\uparrow$), indicating better ability to execute high-fidelity surface manipulation through stable tool orientation, adaptive

What holds

60.5 vs 37.3 (+23.2). Clear baseline matrix incl. π_0 -fast.

Plug story

80 vs 60 vs 45. Naive helps; FVLMoE adds +20.

USB soft

25 vs 5/5/0/0. Force on/off null for both π_0 .

Pump falsifier

Naive w/F 83 > ForceVLA 67. Do not overclaim every column.

Systems/methods preprint — accept these

Avg gap consistent

60.5 vs 37.3 / 40.2 with π_0 -base and π_0 -fast $\pm$ force. Foil set is clear; OpenVLA is citation-only.

Tab.3 causal knife

Early MoE-before-VLM 0% vs late FVLMoE 80%. Design rule: force after VLM; MoE over concat.

Occlusion 90%

Strong multimodal story when vision degrades. Naive force can hurt under shift (base 60 $\rightarrow$ 30).

Honest limitations

Estimated wrench; expensive F/T platforms; accessibility as future work. Related-work separates VLA / contact / MoE.

Grade these without theatrical rejection

Multi-task USB 10%

Tab.5 ForceVLA matches base w/F at 10% USB. Force MoE is not a universal precision-insert fix.

Unstable socket

ForceVLA 20%; π_0 -fast w/F 30% best. Gen claim is not uniform across conditions.

No $\pi_{0.5}$ / OFT bake-off

$\pi_{0.5}$ cited [21] only. No OpenVLA-OFT / 2025 VLA heads. Do not invent a foundation-model ranking.

Small N / one body

~50 demos/task; Flexiv only; Tab.3 task/N underspecified; router Expert-0 role interpretive; no CI/seeds.

Occlusion strong. Unstable soft. Naive can hurt.

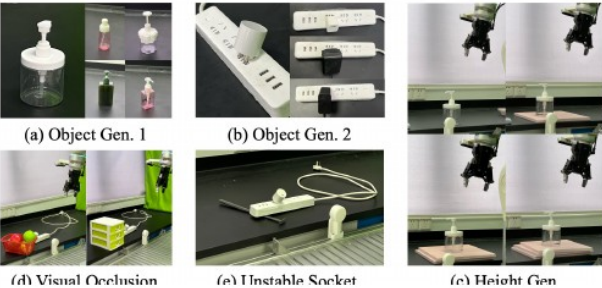
(a) Object Gen. 1 (b) Object Gen. 2 (c) Height Gen.

Figure 6: Variants of generalization settings used in our experiments. (a–b) Different object geometries; (c) variation in socket height; (d) partial visual occlusion; (e) unstable socket conditions. These scenarios evaluate robustness under diverse physical and perceptual perturbations.

Table 2: Success rates (%) of different models under various experimental conditions. Maximum values in each column are highlighted in **bold**; second-best values are underlined.

Model	Object Gen. 1	Object Gen. 2	Height Gen.	Visual Occlusion	Unstable Socket	Average
π_0 -base <u>10</u> w/o F	48.00%	10.00%	66.67%	<u>60.00%</u>	10.00%	38.93%
π_0 -base <u>10</u> w/ F	32.00%	10.00%	77.78%	30.00%	10.00%	31.96%
π_0 -fast <u>25</u> w/o F	80.00%	<u>35.00%</u>	88.89%	50.00%	10.00%	<u>52.78%</u>
π_0 -fast <u>25</u> w/ F	32.00%	5.00%	44.44%	50.00%	30.00%	32.29%
ForceVLA (Ours)	80.00%	40.00%	88.89%	90.00%	<u>20.00%</u>	63.78%

height and measures success under torque limits: (d) **Visual Occlusion**, where parts of the plug and



ForceVLA gen avg

63.78%

Occlusion 90%

Obj Gen.2 40%

Height 88.89% (tie w/ fast w/o F)

Threats

Unstable 20 < fast w/F 30

Naive force lowers both π_0 gen averages.

Joint regime. Plug saturates. USB stays soft.

frequency, reflects not only temporal semantic variance within tasks but also architectural bias toward certain experts, potentially shaped during training.

D Multi-task Evaluation

Table 5: Multi-task joint training success rates (%). ForceVLA (Ours) demonstrates superior average performance and excels or matches the best performance in all individual tasks. Best performance(s) in each column are in **bold**.

Model	Pump Bottle	Insert Plug	Insert USB	Wipe Board	Average SR
π_0 -fast w/o F	0.0%	0.0%	0.0%	0.0%	0.0%
π_0 -fast w/ F	0.0%	0.0%	0.0%	0.0%	0.0%
π_0 -base w/o F	20.0%	0.0%	0.0%	0.0%	5.0%
π_0 -base w/ F	50.0%	100.0%	10.0%	10.0%	42.5%
ForceVLA (Ours)	80.0%	100.0%	10.0%	80.0%	67.5%

ForceVLA avg 67.5

Pump/wipe 80; plug 100; USB 10. Beats base w/F 42.5 and fast 0.

Does not isolate FVLMoE on plug

Base w/F also 100% plug under joint train. USB ties at 10%.

Pump reversal disappears

Fig.5 Pump naive>ForceVLA; Tab.5 Pump ForceVLA 80 > base w/F 50. Regime matters.

π_0 -fast joint collapse

Both fast variants 0% average. Capacity story is plausible, not dissected.

Late force token into FM — what is CLEAR

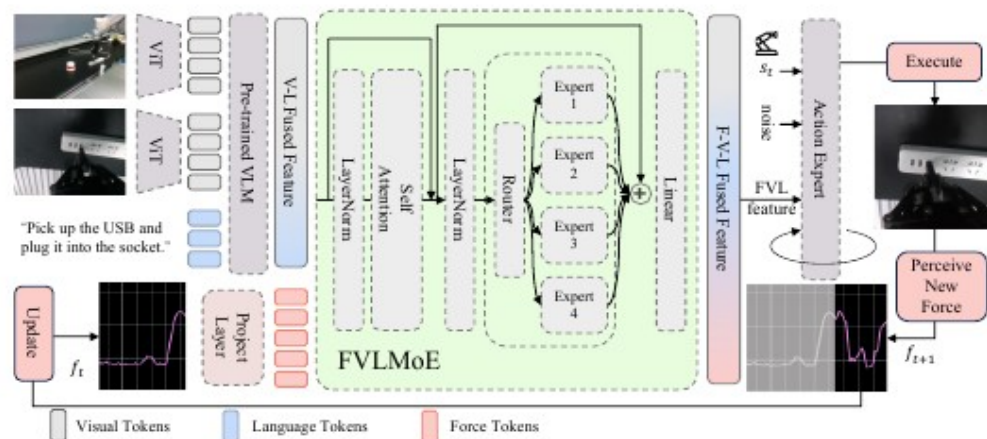


Figure 3: Overview of our ForceVLA model. Visual and language inputs are processed by a pre-trained VLM to form contextual embeddings. External force signals are projected and fused with VLM outputs via the FVLMoE module. The resulting multimodal features guide a flow-based action head to generate contact-aware robot actions.

that are used to select a small subset of k experts (typically $k = 1$ or $k = 2$, where $k \ll N$) from the total pool of N experts. The input token x is then routed only to these k active experts. The outputs of these active experts, $E_i(x)$, are subsequently aggregated, commonly through a weighted sum where the weights $g_i(x)$ are also derived from the gating network. The final output $y(x)$ of the MoE layer can be expressed as: $y(x) = \sum_{i \in \text{TopK}(G(x))} g_i(x) E_i(x)$, where $\text{TopK}(G(x))$ denotes the set of indices of the top- k experts selected by the gating network for input x .

4 ForceVLA

CLEAR

Builds on π_0 FM + PaliGemma. Force after VLM. $E=4$ top-1. Additive into FM suffix.

CLEAR foil

π_0 -base / π_0 -fast $\pm$ force. Not OpenVLA tables.

ABSENT

$\pi_{0.5}$ bake-off. Expert-count sweep. Force-noise ablation. Sim split.

FALSE ancestry

Not MoPA-PD. Not JEPa/WM.

Accept late FVLMoE. Refuse foundation-model upgrade.

Systems/methods preprint for force-augmented VLA fusion.

ACCEPT: late fusion $\gg$ early; FVLMoE $\gg$ naive concat on reported Flexiv contact suite; occlusion story; honest hardware limitations.

WITHHOLD: solved contact-rich foundation model; beats $\pi_{0.5}$; universal insert fix (USB 10% multi-task); uniform gen (unstable socket); causal Expert-0 labels.

Syllabus grade: HIGH for Alternative approaches / contact-rich. Embodiment and open-world: NARROW.

One-liner: Late FVLMoE is the real contribution; naive force is a straw improvement.

Grade the fusion. Keep the numbers honest.

Tab.3 0→80 and +23.2 avg are enough to accept the method claim.

Soft USB, unstable socket, Pump reversal, and missing $\pi_{0.5}$ bake-off keep the paper in Alt-approaches — not foundation coronation.

Next: Empiricist smokes Ablation Bot ranks without reprinting Fig.5 as course smoke.